

Representation, Invariance, and Projection

*A Cross-Disciplinary Theory of What Representations Preserve,
Transform, and Enable*

Formal research paper

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Final determination: 3. A general theory of projection is justified, but semantic invariance requires strict scope conditions.

Abstract

Representation is not one homogeneous relation. Across semiotics, philosophy of science, mathematics, logic, cognitive science, cartography, information visualization, programming languages, databases, compilers, accessibility, and human-computer interaction, a representation is treated variously as a sign relation, a model, a structure-preserving mapping, an encoding, an external cognitive resource, an abstraction, a view, or a public instrument. No single definition subsumes these traditions without loss. A stable cross-disciplinary core nevertheless emerges: a representation is a materially or formally realized vehicle used within an interpretive practice to provide selective access to a declared target under a mapping or rule of correspondence. Representations preserve some relations and observables, transform others, omit others, and introduce operations, saliences, ambiguities, and costs. Equivalence is therefore typed rather than absolute. Syntactic, semantic, informational, structural, topological, computational, inferential, task, perceptual, behavioral, and interactional equivalence impose different identity conditions and require explicit observers, contexts, metrics, and tolerances.

The paper evaluates whether 'projection' adds explanatory value beyond representation. Existing theories supply most component mechanisms: model morphisms, abstraction functions, database views, bidirectional lenses, encodings, simulations, and scientific models. They do not provide one shared contract linking preservation and loss to epistemic access and action opportunity across formal and empirical domains. Projection is therefore justified as a cross-disciplinary analytical role, not as a new primitive substance: a declared transformation from a target description to a representation form for specified observers and uses. Semantic invariance is coherent only when the target content, interpretation rules, admissible contexts, observables, and tolerance are fixed. Informational equivalence does not imply inferential, perceptual, task, behavioral, or interactional equivalence. The resulting reusable artifacts are a representation taxonomy, typed equivalence matrix, projection contract, comparison protocol, and falsification matrix.

Primary contribution: representation change can be analyzed rigorously by declaring a target, mapping, preservation profile, loss profile, operations, observer model, use conditions, and failure conditions; no unqualified semantic-invariance claim is scientifically defensible.

1. Problem Statement

Scientific and computational work depends on representations while repeatedly warning that representations are not their targets. Equations, diagrams, data tables, maps, simulations, source code, database views, interfaces, and narratives can concern the same target yet support different inferences and actions. The central problem is not merely that formats differ. It is that every representation selects structure, establishes observables, redistributes cognitive work, and constrains what can be inspected or changed.

Three errors follow from treating representation as transparent. First, identity is inferred from common reference: two vehicles are assumed equivalent because they concern the same target. Second, meaning is inferred from information preservation: equal recoverable data are assumed to yield equal understanding. Third, behavior is attributed directly to representation without specifying observers, tasks, prior knowledge, context, or available operations. This paper rejects all three inferences.

The research question is: What is a representation, what can remain invariant across representations, and what changes when the same underlying target is represented differently? The inquiry is

independent of specific interfaces and software architectures. It asks whether existing theories are sufficient or whether a general theory of projection is required to coordinate them.

2. Literature Review Method

The review uses a structured conceptual synthesis rather than a statistical meta-analysis. It samples foundational and primary work from philosophy of science and language, semiotics, formal logic, mathematics, scientific modeling, information theory, cognitive science, distributed and external cognition, visualization, cartography, programming-language semantics, compilers, databases, knowledge representation, accessibility, and HCI. Sources were selected when they explicitly define representation, specify a mapping or interpretation relation, characterize preservation or loss, define an equivalence relation, or report effects of representational form on reasoning and action.

Claims were coded along eight dimensions: vehicle, target, mapping, interpreter, preserved properties, transformed or omitted properties, available operations, and evaluation criterion. Contradictory positions were retained. The synthesis does not use a pre-existing interaction ontology as evidence. It distinguishes observations reported in empirical studies from philosophical interpretation and from the paper's proposed analytical framework.

Limitations are material. Several traditions use the same word for different roles, while others avoid the word but formalize related structures. Formal results establish properties relative to stipulated semantics and observables; they do not automatically generalize to human interpretation. Empirical results establish performance effects under specific tasks and populations; they do not prove universal representational mechanisms.

3. Major Theories of Representation

3.1 Semiotic and referential accounts

Peircean semiotics treats a sign through a triadic relation among sign, object, and interpretant. Signhood is relational rather than an intrinsic property of a vehicle. This blocks a target-only definition: a mark is not a representation merely by resembling or correlating with something; it functions representationally through an interpretive relation. Saussurean traditions emphasize differences within a sign system and conventional relations between signifier and signified. Goodman rejects resemblance as a necessary basis of representation and analyzes symbol systems through reference, denotation, exemplification, and syntactic and semantic differentiation.

Strength: these accounts explain why identical vehicles may mean differently across practices and why dissimilar vehicles may represent the same target. Limit: they offer limited quantitative guidance about preserved structure, computational behavior, or task performance. Their central constraint is indispensable: representation requires a rule-governed relation, not mere physical similarity.

3.2 Philosophy of science and scientific modeling

Scientific-representation theories disagree over whether representation is resemblance, isomorphism, partial isomorphism, similarity, inference, or use. Structural accounts emphasize mappings between model structure and target structure. Similarity accounts make representation selective and purpose-relative. Inferential conceptions emphasize a model's capacity to license competent reasoning about a target. Pragmatic accounts place agents, purposes, and contexts inside the representation relation.

Perspectival approaches stress that scientific representations disclose selected aspects under conditions rather than mirror complete reality.

A common result is negative: no representation represents everything about its target. Scientific models idealize, abstract, distort, and sometimes deliberately misrepresent while remaining epistemically useful. Accuracy is multidimensional and purpose-relative, but not arbitrary; empirical fit, robustness, inferential reliability, and error characterization constrain acceptable use.

3.3 Mental, external, and distributed representation

Cognitive science uses 'representation' for internal states or structures that carry content and participate in computation, but representationalism is contested by ecological, enactive, and dynamical approaches. External-cognition research shifts attention from internal content to the joint system formed by agents and external structures. Zhang and Norman's representational analysis shows that formally equivalent task structures can distribute information differently between mind and environment, changing memory demands and permissible operations. Hutchins extends the unit to socially and materially distributed cognitive systems.

Strength: these traditions explain why a vehicle can reorganize reasoning without changing target information. Limit: claims about internal representations remain theoretically disputed, and external performance effects depend on learned conventions, perceptual capacities, and task structure.

3.4 Mathematics, logic, and model theory

Mathematics supplies precise preservation vocabulary. A homomorphism preserves specified operations or relations; an isomorphism is a structure-preserving bijection with an inverse of the relevant kind; homeomorphism preserves topological structure; equivalence relations partition objects under declared criteria. Model theory distinguishes a formal language, a structure interpreting it, and satisfaction. Logical equivalence concerns truth across interpretations; elementary equivalence concerns satisfaction of the same first-order sentences; definitional and categorical equivalences impose different translations between theories.

These results establish that invariance is always indexed to structure. Isomorphic objects may differ physically; homeomorphic spaces may differ metrically; logically equivalent expressions may differ cognitively or computationally. Formal equivalence cannot be transferred across levels without proof.

3.5 Information theory and coding

Information theory formalizes uncertainty reduction, channel capacity, coding, and loss without supplying semantic meaning. Shannon explicitly brackets semantic aspects of communication from the engineering problem. Lossless coding permits exact reconstruction of a source sequence under a code; lossy coding permits controlled distortion under a measure. Sufficient statistics preserve all information in a sample relevant to a specified parameter, demonstrating again that preservation is relative to an inferential target.

Strength: exact measures of recoverability, uncertainty, and distortion. Limit: identical information-theoretic content can support different interpretations, inferences, and actions. Semantic or behavioral equivalence cannot be inferred from entropy or decodability alone.

3.6 Cartography and information visualization

Cartography treats projection explicitly as a mapping from one geometric surface to another. No flat world map preserves area, angle, shape, direction, and distance globally. Projection families declare which properties are preserved and where distortion occurs. Visualization similarly encodes data

attributes into graphical marks and channels. Graphical-perception experiments show that encodings differ in decoding accuracy; position along a common scale generally supports more accurate quantitative judgment than area or volume. A visualization may therefore be data-correct while inferentially poor.

These fields provide the clearest empirical analogy for a projection contract: source domain, target form, invariant class, distortion class, and intended use must be stated together.

3.7 Programming languages, compilers, and computation

Programming-language semantics distinguishes syntax from denotation, operational behavior, and observations. Two programs may be syntactically different yet denotationally equivalent, contextually equivalent, or bisimilar under defined observations. Compiler correctness is commonly expressed as preservation of semantics or observable behavior from source to target, with explicit allowances for undefined behavior, resource variation, and implementation-specific observations.

Abstract interpretation formalizes sound approximation between concrete and abstract semantic domains, often through Galois connections. Soundness preserves truth of specified properties from abstraction to concrete behavior; completeness is stronger and frequently unavailable. This is a mature theory of projection-like mapping, but its scope is formal semantic analysis rather than human epistemic access.

3.8 Databases, views, and bidirectional transformations

Codd's relational model separates logical data structure from physical representation and motivates data independence. A database view exposes a derived representation of source data. View-update theory asks how changes to the view correspond to changes in the source. Bidirectional transformations and lenses formalize paired mappings subject to round-trip laws. These theories demonstrate that projection is not only read access: available updates and their consistency conditions are part of the representational relation.

Limit: database equivalence normally presumes formal schemas and well-defined queries. It does not explain salience, discoverability, perceptual interpretation, or learned action strategies.

3.9 Knowledge representation

Knowledge representation treats formal languages as surrogates for domains, commitments about what exists, fragments of reasoning, media for computation, and means of human expression. Ontologies, description logics, frames, semantic networks, and probabilistic models preserve different inferential commitments. Expressiveness trades off against tractability and decidability. Two formalisms may encode corresponding facts yet differ in what can be inferred efficiently or transparently.

3.10 Accessibility, multimodality, and HCI

Accessibility standards distinguish content and relationships from sensory presentation. Requirements not to depend solely on color, shape, location, or sound imply that a target instruction or relation should survive modality change. Yet modality substitution is not identity: spatial overview, temporal scanning, privacy, precision, fatigue, and learnability can differ. HCI adds mappings, feedback, signifiers, constraints, and action availability. Interface forms are therefore not passive displays of an invariant semantics; they mediate observation and intervention.

4. Representation-Target Relations

A representation cannot be adequately defined by a vehicle alone. The minimum relation includes a vehicle V , a target T , a correspondence or construction rule M , an interpretive practice I , and a use context C . The target need not be physical reality. It may be a scientific object, dataset, formal structure, process, model, possible state, rule system, or prior representation. Nor must the target be fully determinate independent of practice: classifications and model-defined quantities may be partly constituted by the representational framework.

The paper adopts the following neutral definition: A representation is a materially or formally realized vehicle that, within an interpretive practice, provides selective access to a declared target by means of a rule-governed correspondence. 'Selective access' covers reference, measurement, inference, simulation, communication, coordination, and intervention. The definition does not require resemblance, mental content, full fidelity, or observer-independent semantics.

Meaning, content, information, and interpretation remain distinct. Information concerns distinctions and recoverability under a code or probability model. Content is what a representation is taken to be about or to assert under a semantics. Meaning includes inferential, pragmatic, or use consequences within an interpretive system. Interpretation is the process or rule by which a vehicle is related to content or target. A single information-bearing pattern can receive multiple interpretations; a common content can be encoded with different information costs.

5. Preservation, Transformation, and Loss

Representation is selective preservation under transformation. Preservation claims identify a relation or observable P such that relevant distinctions in the target remain recoverable or valid in the representation. Transformation claims identify systematic changes to form, scale, granularity, coordinate system, ordering, or dynamics. Omission claims identify target distinctions not encoded or not recoverable. Introduction claims identify structure contributed by the representation, including categories, visual grouping, defaults, coordinate axes, interaction operations, or uncertainty not present in the target description.

Property	Meaning of preservation	Typical test	Failure example
Identity	Declared target entities remain distinguishable under the mapping.	Injectivity or stable identifiers under scope.	Aggregation merges distinct entities.
Semantic relations	Declared propositions or reference relations retain truth conditions.	Interpret both forms under a common semantics.	A label translation changes category boundaries.
Structure	Specified relations and operations commute with the mapping.	Homomorphism/isomorphism test.	A table omits graph reachability structure.
Topology	Continuity, adjacency, or neighborhood is retained.	Homeomorphism or adjacency comparison.	Map projection tears or changes neighborhood.
Order	Ordering relations remain valid.	Monotonicity/order isomorphism.	Alphabetical sorting destroys temporal sequence.
Quantity	Specified measures remain accurate within tolerance.	Metric error bounds.	Area encoding biases magnitude estimates.
Temporal structure	Ordering, duration, or simultaneity is preserved.	Trace or temporal-logic comparison.	Narrative linearizes concurrent events.
Uncertainty	Probability, interval, provenance,	Calibration and propagation	Point estimate suppresses

Property	Meaning of preservation	Typical test	Failure example
	or confidence is retained.	tests.	distribution.
Constraints	Valid and invalid states/actions remain correspondingly represented.	Constraint satisfaction or round-trip tests.	View admits updates with no valid source update.
Computational behavior	Declared observations of execution remain the same.	Contextual equivalence, bisimulation, refinement.	Optimization changes floating-point results.
Inferential consequences	Queries or conclusions in a declared logic are preserved.	Theory interpretation or query equivalence.	Abstraction proves only a strict subset.
Action possibilities	Specified feasible operations and effects correspond.	Operation-set and transition comparison.	Read-only view removes update capability.

Loss cannot generally be measured independently of purpose and observer. Information-theoretic loss requires a source distribution and distortion measure. Structural loss requires a declared signature. Semantic loss requires an interpretation. Task loss requires a performance criterion. Perceptual loss requires an observer model. A purpose-independent catalogue of omitted distinctions is possible, but assigning importance to those omissions is not.

6. Typed Equivalence Relations

'Equivalent' is scientifically incomplete unless the equivalence type, observables, context, and tolerance are stated. The following matrix distinguishes relations that are often conflated.

Type	Must remain the same	May differ	Test / scope condition
Syntactic	Form under a grammar or normalization rule.	Meaning, behavior, use.	Token/tree equality or canonicalization.
Semantic	Denotation or truth conditions under a declared semantics.	Syntax, cost, salience, interaction.	Interpretation equality over specified models.
Informational	Recoverable distinctions or mutual reconstructability.	Presentation, inference cost, behavior.	Lossless transforms or bounded information loss.
Structural	Selected relations and operations.	Unselected properties and material realization.	Homomorphism/isomorphism over a signature.
Topological	Continuity and neighborhood relations.	Metric, angle, area, scale.	Homeomorphism or topology-preserving map.
Computational	Declared observable execution behavior.	Internal steps, resources, syntax.	Contextual equivalence, bisimulation, refinement.
Inferential	Answers or consequences for a specified query/logic class.	Other consequences, cognitive cost.	Query equivalence or consequence preservation.
Task	Success distribution for a specified task and population.	Strategies, time, errors outside metric.	Equivalence/non-inferiority with tolerance.
Perceptual	Discriminability or perceptual organization under conditions.	Semantics and task outcomes.	Psychophysical discrimination thresholds.
Behavioral	Distribution of declared actions/outcomes.	Internal interpretation and unmeasured behavior.	Controlled comparison over population/context.
Interactional	Available operation-transition-observation structure.	Surface form and internal implementation.	Bisimulation-like comparison plus human-use tests.

Known counterexamples show non-entailment among types. Algebraically equivalent equations can produce different solution strategies. Informationally equivalent graphs and tables can differ in comparison accuracy. Semantically equivalent programs can differ in time, energy, numerical stability,

or security side channels. Equal task completion can conceal different understanding, error recovery, and transfer. Behavioral equivalence under one context can disappear after learning or under time pressure.

7. Invariance Under Projection

An unrestricted claim that meaning remains invariant under representation change is too strong. 'Same target' does not guarantee same content, because a projection may alter categorization, precision, presupposition, indexical frame, or available interpretation. A coherent invariance claim must specify at least six items: the target description; the content or property family asserted invariant; the mapping between forms; the interpretation rules; the admissible contexts and observers; and the metric or tolerance.

Let T be a declared target domain, R_1 and R_2 representation domains, $p_1:T \rightarrow R_1$ and $p_2:T \rightarrow R_2$ projection functions, and J_1 and J_2 interpretation functions into a comparison domain K . Content invariance for property family Q holds when $J_1(p_1(t))$ and $J_2(p_2(t))$ agree on Q for all t within the declared domain and tolerance. This is not circular if Q and the interpretation functions are independently specified. It becomes circular when equivalence is defined merely as 'preserving the same meaning' and meaning is inferred only from judged equivalence.

Target invariance, content invariance, meaning invariance, behavioral invariance, task-outcome invariance, and interpretation invariance are independent claims. A target can remain fixed while interpretations vary. Content can be logically invariant while behavior changes. Task outcomes can remain equal while strategies and understanding differ. Interpretation can converge while feasible actions differ.

The evidence therefore supports the phrase 'scoped semantic invariance,' not semantic invariance simpliciter. Formalization is possible where target semantics and correspondence rules are explicit. Empirical testing is required where interpretation depends on human observers, learned conventions, modality, or context.

8. Epistemic Access and Opportunity

Representations actively organize epistemic access. They determine which distinctions are visible without transformation, which comparisons are local, what must be remembered, what can be searched, what errors become salient, and which inferences are executable through perceptual or computational operations. External representations can convert internal computation into perceptual recognition, but they can also conceal provenance or encourage invalid inferences.

Opportunity should not be introduced as a replacement for affordance, choice set, action space, or information access. Existing concepts cover much of the terrain. Ecological affordances concern action possibilities relative to organism-environment systems. Decision theory and reinforcement learning formalize available actions and policies. Control theory formalizes reachable states and controllability. HCI distinguishes actual, perceived, and signaled actions. Visualization and external cognition analyze operations and inferential access.

A cross-disciplinary analysis nevertheless requires three distinct sets: feasible actions F , signaled or directly discoverable actions S , and believed actions B for an observer in context. Representational change may leave F fixed while changing S and B ; it may also change F when the representation is an

operative view that permits or blocks transformations. 'Opportunity profile' is retained only as a descriptive tuple $\langle F, S, B, \text{inferential operations}, \text{error-recovery operations}, \text{coordination operations} \rangle$, not as a new ontological entity.

Representation does not cause behavior directly. A defensible causal model includes representation, observer capabilities and prior knowledge, task, incentives, environment, available operations, and temporal history. Experimental attribution requires manipulation of representation while controlling or measuring these mediators and moderators.

9. Comparative Cases

9.1 One semantic structure as table, graph, map, and narrative

Target: a directed network of public-health facilities, referral relations, service capacities, locations, and temporal disruptions. A table preserves attribute values and supports sorting, filtering, and exact lookup; it suppresses path structure unless computed. A node-link graph exposes connectivity and centrality but can obscure exact quantities and geographic distance. A map preserves selected spatial relations and supports regional comparison while distorting geometry according to projection and potentially hiding non-spatial dependencies. A narrative preserves causal or temporal interpretation and supports explanation but linearizes concurrency and makes exhaustive comparison difficult.

The target need not change, but no pair is globally equivalent. They may be informationally equivalent if each can be reconstructed from the same dataset, structurally equivalent for selected relations, or task-equivalent for a specific query. Behavioral predictions include faster exact lookup in tables, faster path recognition in graphs, faster spatial clustering judgments in maps, and better causal recall from narrative; these remain empirical hypotheses conditioned on scale, literacy, and task.

9.2 One scientific target through equations, diagrams, simulation, and prose

Target: a susceptible-infectious-recovered epidemic model with specified parameters and assumptions. Differential equations preserve formal dependencies and permit analytic reasoning. A compartment diagram exposes flows and conservation structure. Simulation exposes trajectories, stochastic variation, and sensitivity under parameter settings. Prose states assumptions, interpretations, and limitations not contained in the equations. The forms are complementary rather than substitutable. Equation and code may disagree through discretization or implementation error; a diagram may omit nonlinear terms; prose may introduce ambiguity. Scientific adequacy requires cross-representation consistency tests.

9.3 One computational system through GUI, CLI, API, and conversation

Target: a version-control system's repository states and operations. A GUI externalizes status and constrains commands, supporting recognition and discoverability. A command line provides compositional, scriptable operations and exact textual traces but requires recall. An API exposes machine-addressable operations, schemas, and errors while omitting interactive guidance. A conversational interface maps natural-language requests to operations, lowering formulation cost but adding interpretive uncertainty and a need for confirmation and provenance.

Core repository transitions may be computationally equivalent when each invokes the same validated operations. They are not interactionally equivalent: available compositions, visibility of consequences, error prevention, and auditability differ. Equal task completion does not establish equal understanding or safety.

9.4 Accessibility across visual, auditory, and tactile forms

Target: a transit network with stations, lines, transfer points, service alerts, and route options. A visual map offers parallel spatial overview. Auditory presentation serializes access and increases memory and navigation costs unless structured for query. A tactile map preserves selected topology and spatial layout but has resolution and labeling constraints. Multimodal forms can preserve route semantics while differing in perceptual access, privacy, scanning time, and error recovery. Accessibility requires content and relations to be programmatically determinable or available in alternative forms, but equivalent access is a performance and usability claim, not a consequence of nominal content duplication.

9.5 Informational equivalence with different reasoning

The classic isomorphic arithmetic-task results analyzed by Zhang and Norman show that representations distributing information externally can change problem difficulty despite equivalent abstract structure. Visualization research similarly shows that equal numerical data encoded by position, length, angle, area, or color are decoded with different accuracy. These cases falsify the inference from informational equivalence to cognitive or behavioral equivalence.

10. Candidate Formal Model

Projection is defined here as an analytical role: a declared transformation that produces a representation of a target for specified observers and uses. It does not replace 'representation,' 'encoding,' 'view,' 'abstraction,' 'model,' or 'simulation' inside disciplines where those terms are sufficient. It supplies a common comparison schema across them.

A projection contract is the tuple $\Pi = \langle T, R, p, I, P, X, O, A, U, C, E \rangle$, where T is the source or target domain; R is the representation domain; p is the transformation or construction rule; I is the interpretation rule or practice; P is the preservation profile; X is the transformation, omission, and introduction profile; O is the observer model; A is the available operation set; U is intended use; C is context and scope; and E is error, uncertainty, tolerance, and failure conditions.

For a target state t , the representation $r=p(t)$ is scientifically analyzable when the contract permits independent investigators to determine what claims are licensed, which distinctions are unavailable, how operations correspond to target transitions, and under what conditions conclusions fail. The model is deliberately weaker than a universal semantics. It allows formal mappings, physical instruments, public symbols, and learned representational practices.

A comparison between projections Π_1 and Π_2 is made over a declared property family Q and observer-task distribution D . Formal preservation is tested by commuting diagrams, reconstruction, logical consequence, or behavioral refinement where applicable. Empirical access is tested through accuracy, latency, search paths, confidence, error types, learning, transfer, and recovery. Causal claims require randomized or otherwise identified manipulation and mediation analysis where feasible.

11. Falsification Tests

Claim	Evidence against	Consequence
Representation requires a target and interpretive rule.	Stable predictive representation relation definable solely by intrinsic vehicle properties.	Revise relational definition.
Equivalence must be typed.	One equivalence relation proven to preserve	Collapse taxonomy.

Claim	Evidence against	Consequence
	all scientifically relevant properties across domains.	
Informational equivalence does not imply behavioral equivalence.	Controlled studies consistently show identical action distributions across informationally equivalent forms and populations.	Remove behavioral separation.
Scoped semantic invariance is coherent.	No non-circular independent specification of content and interpretation can be constructed even in formal cases.	Reject semantic-invariance language.
Projection adds cross-disciplinary value.	Existing single framework already covers formal preservation, human epistemic access, operations, uncertainty, and failure without translation loss.	Use that framework; discard projection contract.
Representations organize epistemic access.	After controlling task, observer, operations, and context, representational manipulation produces no reliable differences in access measures.	Restrict claim to formal access only.
Opportunity profile is not reducible to one existing construct.	Affordance, action-space, or view theory fully captures feasible, signaled, believed, inferential, recovery, and coordination operations.	Adopt established construct exclusively.
Representation and implementation can often be separated.	Preservation and operation claims cannot be stated independently of device, rendering, data structure, or execution substrate in target cases.	Treat separation as domain-limited.

The most damaging result would be demonstration that the projection contract merely restates an existing general theory with no added comparative predictions or measurement discipline. The framework therefore earns retention only if it improves cross-form specification, reveals hidden non-equivalences, or supports reproducible comparative experiments.

12. Threats to Validity

Construct validity is threatened by polysemy. 'Representation,' 'model,' 'meaning,' 'information,' and 'equivalence' change across traditions. The paper mitigates this by defining roles and typed relations rather than asserting lexical identity. External validity is threatened by moving from formal domains with stipulated semantics to human contexts with negotiated interpretation. Formal claims are therefore not generalized without observer and context variables.

Selection bias remains because the literature is too large for exhaustive review. The synthesis favors foundational accounts and explicit formalizations, potentially underrepresenting practice-based, non-Western, phenomenological, and artistic theories. Empirical claims in visualization and cognition often use restricted tasks and participant populations. Accessibility equivalence is especially sensitive to heterogeneous capabilities and technologies.

Causal validity is limited. Observed differences between representations may arise from familiarity, training, ordering, modality, incentives, or implementation quality. Studies must control content and available operations while avoiding the false assumption that surface matching controls interpretation. Finally, the proposed contract may be too general to generate novel predictions; its value should be judged by comparative use, not conceptual elegance.

13. Implications for Interaction Research

Interaction research should not posit an observer-independent semantic state merely because multiple interfaces address a common backend object. It should declare the target description and authority that stabilizes it. Representation, opportunity, interpretation, action, and outcome should remain separate variables. Feasible actions may be invariant while signaled and believed actions vary; identical actions may be interpreted differently; equal outcomes may arise through different strategies.

Representation similarity should be typed. A GUI and API may be computationally equivalent but perceptually and interactionally dissimilar. Behavioral equivalence requires a population, context, task distribution, metrics, and tolerance. Transfer between representations should be analyzed through preserved conceptual relations, learned operations, and changed cues rather than through visual resemblance alone.

The projection contract can support experiments by making manipulation explicit. Researchers can hold target and task constant, vary one projection property, measure epistemic access and opportunity profiles, and test mediators such as salience, memory load, inferred action set, or confidence. This creates falsifiable work without presuming a new discipline or ontology.

14. Conclusion

Cross-disciplinary evidence rejects a unitary theory in which representation is resemblance, encoding, formal mapping, mental state, or public artifact alone. It also rejects absolute equivalence and unrestricted semantic invariance. Representations are selective, rule-governed means of access to declared targets within interpretive practices. They preserve some distinctions, transform or omit others, and introduce operations and organizational effects.

Existing theories are powerful but fragmented by domain. Model theory, morphisms, information theory, abstract interpretation, observational equivalence, database views, lenses, cartographic projection, semiotics, scientific modeling, and external cognition already explain major components. No evidence supports replacing them with a wholly new representation theory. A general theory of projection is justified only as a shared analytical contract that coordinates those established mechanisms and requires explicit scope conditions.

Final determination: 3. A general theory of projection is justified, but semantic invariance requires strict scope conditions.

15. Reusable Representation-Analysis Framework

A. Representation Taxonomy

Type	Defining relation	Primary preservation concern	Typical risk
Indexical / measurement	Causal or calibrated relation to target.	Traceability, quantity, uncertainty.	Instrument bias or invalid calibration.
Symbolic / linguistic	Conventional sign system and interpretation.	Reference and inferential role.	Ambiguity and context shift.
Structural / mathematical	Mapping between relational structures.	Selected relations and operations.	Mistaking partial for total isomorphism.
Model / scientific	Purposeful surrogate used for inference.	Empirical and explanatory relevance.	Idealization outside scope.
Diagrammatic / visual	Spatial encoding of relations or quantities.	Topology, order, magnitude, grouping.	Perceptual distortion or salience bias.
Narrative	Temporally and causally organized discourse.	Sequence, agency, explanation.	Linearization and selective framing.
Encoding / code	Rule-based transformation of symbols or signals.	Decodability and error bounds.	Semantic indeterminacy.
Abstraction	Sound approximation of a richer domain.	Specified properties or judgments.	False completeness.
View / projection	Derived exposure of source structure.	Query answers and update correspondence.	Hidden source distinctions.
Simulation	Executable model producing trajectories.	Dynamic behavior under assumptions.	Implementation and parameter artifacts.
Interface / operative representation	Observable and manipulable access surface.	Operation-transition-observation relation.	Confusing discoverability with feasibility.
Internal cognitive	State posited to carry content and guide processing.	Explanatory role in cognition.	Unobservable or theory-dependent content.

B. Equivalence Matrix

Use the typed equivalence table in Section 6. Every equivalence claim must complete: type; invariant; admissible difference; observer; task; context; metric; tolerance; and test. Multiple types may be asserted, but none may be inferred from another without evidence.

C. Projection Contract

Field	Required declaration
Target domain	What is represented; authority and version; boundaries and granularity.
Representation domain	Vehicle/form and admissible instances.
Mapping	Construction, encoding, measurement, query, or transformation rule.
Interpretation	Semantics, conventions, training, and decoding assumptions.
Preserved properties	Relations, quantities, behaviors, consequences, or operations retained.
Transformed properties	Changes of coordinate, scale, order, granularity, modality, or dynamics.
Omitted properties	Distinctions unavailable or unrecoverable.
Introduced properties	Categories, saliences, defaults, ambiguity, artifacts, or noise.
Available operations	Read, query, compare, manipulate, update, undo, coordinate, export.
Observer assumptions	Capabilities, knowledge, language, access needs, and role.

Field	Required declaration
Intended use	Questions, decisions, tasks, and prohibited uses.
Uncertainty and error	Provenance, confidence, distortion, latency, and tolerance.
Failure conditions	Known counterexamples, out-of-scope states, and invalid inferences.

D. Representation Comparison Protocol

1. Declare one target and its authoritative boundaries without assuming observer-independent meaning.
2. Inventory each representation vehicle, mapping, interpretation rule, and available operation.
3. Specify candidate invariants and the equivalence type for each; reject untyped equivalence claims.
4. Construct preservation, transformation, omission, and introduction profiles.
5. Hold target content and task constant where possible; record unavoidable implementation differences.
6. Declare observer population, prior knowledge, modality, context, and accessibility conditions.
7. Measure formal correctness separately from epistemic access, task performance, behavior, learning, and transfer.
8. Test operation correspondence, error recovery, uncertainty communication, and update semantics.
9. Analyze mediation and moderation rather than attributing behavior directly to representation.
10. Report scope, tolerances, contradictory results, and conditions under which equivalence fails.

E. Falsification Matrix

Use the matrix in Section 11 as the minimum test set. Any application should add domain-specific failure conditions and identify which result would require abandoning, narrowing, or replacing the projection analysis.

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